Hyperlocal Courier Dispatch & Tracking Engine

Requirements Table and Use-Case Flow Specification

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall match an outgoing courier request with the nearest active delivery rider within a 3 km radius.	High	Pass: Rider receives dispatch notification with pickup location; Fail: Order assigned to offline rider.	Ensures quick response times and efficient resource allocation.
FR-002	Functional	The system shall allow a sender client to create a courier request with pickup/delivery addresses, package details, and preferred pickup time.	Medium	Pass: Request is created with all fields and a confirmation ID is generated; Fail: Required fields missing causing validation error.	Core feature enabling order placement.
FR-003	Functional	The system shall allow a delivery rider to view available pickup requests and accept one.	High	Pass: Rider sees list of pending requests and accepts one; Fail: Rider cannot accept if already assigned to another.	Enables rider engagement and order assignment.
FR-004	Functional	The system shall generate an optimal multi-stop route for a rider who has accepted multiple deliveries.	Medium	Pass: Route is displayed with ordered stops and estimated times; Fail: Route generation fails or ignores constraints.	Improves delivery efficiency and reduces travel time.
FR-005	Functional	The system shall enforce OTP verification at the delivery destination before completing the delivery.	High	Pass: Delivery completes only after valid OTP is entered; Fail: Delivery marked as complete without OTP.	Ensures secure handover and prevents fraud.
NFR-001	Non-functional (Performance)	Delivery status and live GPS telemetry updates from riders shall transmit to the sender client with under 2-second latency.	High	Pass: Benchmarking tests confirm the target latency under simulated peak load.	Provides real-time visibility to senders, improving trust.
NFR-002	Non-functional (Reliability)	The system shall ensure 99.9% uptime during peak business hours (8 AM – 10 PM).	High	Pass: Monitoring logs show uptime ≥ 99.9% over a one-month period; Fail: Uptime drops below threshold.	Critical for business operations; downtime leads to revenue loss and customer dissatisfaction.

2. UML Use-Case Diagram Reference

System: Hyperlocal Courier Dispatch & Tracking Engine

Actors: Sender Client, Delivery Rider, Support Agent

Use Case ID	Use Case	Primary Actor	Related Requirement
UC-01	Create Courier Request	Sender Client	FR-002
UC-02	Track Delivery	Sender Client	NFR-001
UC-03	Accept Delivery	Delivery Rider	FR-003
UC-04	Complete Delivery	Delivery Rider	FR-005
UC-05	Generate Multi-Stop Route	Delivery Rider	FR-004
UC-06	View Available Requests	Delivery Rider	FR-003
UC-07	Match Nearest Rider	System	FR-001
UC-08	Verify OTP	System / Delivery Rider	FR-005
UC-09	Handle Invalid OTP / Manual Review	Support Agent	FR-005

Relationships represented in the diagram: UC-01 «include» UC-07; UC-04 «include» UC-08; UC-09 «extend» UC-08 when OTP validation fails after the allowed attempts; UC-05 «extend» UC-03 when multiple deliveries require route optimization.

3. Use-Case Flow Specification: Complete Delivery (UC-04)

Actor: Delivery Rider

Preconditions:

- The rider has accepted the delivery request.
- The rider is physically at the delivery destination.

Postconditions:

- Delivery is marked as completed in the system.
- OTP is validated and stored.
- Sender client receives a delivery confirmation notification.

Main Success Scenario:

1. Rider opens the delivery order on the mobile app.
2. System displays the delivery details and a prompt to enter OTP.
3. Rider requests the recipient for the OTP received via SMS/email.
4. Rider enters the OTP into the application.
5. System validates the OTP against the stored value for this delivery.
6. System updates the delivery status to 'Completed'.
7. System sends a confirmation notification to the sender client (e.g., push notification or email).
8. The use case ends successfully.

Alternate Flow – OTP Incorrect:

5a. System detects that the entered OTP does not match the stored value.

5a1. System displays an error message: "Invalid OTP. Please try again."

5a2. Rider is given up to 3 attempts to enter the correct OTP.

5a3. If the rider succeeds within 3 attempts, the flow resumes at Step 6.

5a4. If all 3 attempts fail, the system locks the delivery and flags it for manual review by a support agent.

5a5. The use case ends with a failure status, and the rider is instructed to contact support.

Usecase diagram:

